

# LandslideSOS Backend - Error & Issue Log

Compiled 2026-09-15 after the full audit ('check all updates') and the subsequent fixes. Covers runtime/environment failures, CRITICAL/HIGH/MEDIUM/LOW audit findings, and the verification results after remediation.

## 1. Runtime & Environment Failures

| #  | Error                                                                 | Root cause                                                                            | Fix                                                                                   |
|----|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| 1  | ImportError: DLL load failed: The specified module could not be found | Windows AppCompat shim policy; heartbeats loaded by CPython runtime; fonts            | Reinstall CPython runtime; fonts; switched to native XGBoost                          |
| 2  | AttributeError: module 'xgboost' has no attribute 'DMatrix'           | xgboost wrapper uses xgb.DMatrix instead of booster API                               | xgboost wrapper uses xgb.DMatrix instead of booster API                               |
| 3  | KeyError: 'db'                                                        | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 4  | KeyError: 'sqlalchemy'                                                | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 5  | KeyError: 'celery'                                                    | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 6  | KeyError: 'httpx'                                                     | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 7  | KeyError: 'smoke'                                                     | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 8  | KeyError: 'indention'                                                 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 9  | KeyError: 'imd'                                                       | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |
| 10 | KeyError: 'verbose'                                                   | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 | on sqlalchemy-sqlalchemy-sqlite://localhost:30000/sqlalchemy-sqlite://localhost:30000 |

## 2. Audit Findings (severity CRITICAL/HIGH)

### CRITICAL - #1

**Issue:** Initial migration 7702ae8aba53 creates alerts.status CHECK/Enum constraint with only (pending, delivered, acknowledged). Model adds AlertStatus.failed. On a DB built purely from alembic upgrade head, writing failed violates the CHECK constraint.

**Resolution:** FIXED - added 'failed' to the enum in 7702ae8aba53. Verified: wrote AlertStatus.failed with sms\_failed=1 successfully on a fresh migration-built DB.

### CRITICAL - #2

**Issue:** Phase-3 migration 99f87406c0b8 adds alerts.sms\_failed as NOT NULL with NO server\_default. Fails on existing rows in both SQLite and PostgreSQL.

**Resolution:** FIXED - server\_default='0' added. Pristine upgrade head now runs clean to head.

### CRITICAL - #3

**Issue:** Phase-3 migration 99f87406c0b8 creates the triggered\_by FK with name=None; downgrade calls op.drop\_constraint(None, 'alerts', type\_='foreignkey') which cannot render - broken downgrade path.

**Resolution:** FIXED - explicit fk\_alerts\_triggered\_by on both create (upgrade) and drop (downgrade).

### CRITICAL - #4

**Issue:** predict\_static\_proba always builds an xgb.DMatrix and calls booster.predict. A legacy RandomForest model (sklearn pickle) would raise AttributeError.

**Resolution:** FIXED - branch on bundle['model\_type']: xgboost uses DMatrix/booster, else predict\_proba(fv)[0][1].

### HIGH - #5

**Issue:** `_parse_date` in `inventory/loader.py` looped over formats but ALWAYS called `date.fromisoformat(val.replace('/', '-'))` - the format strings were ignored, so `yy/mm/dd` vs `dd/mm/yy` could be mis-parsed.

**Resolution:** FIXED - uses `datetime.strptime(val, fmt)` for each declared format.

**HIGH - #6**

**Issue:** `sms_failed` column on production PostgreSQL (unrunnable as-is) - covered by CRITICAL #2.

**Resolution:** FIXED together with #2.

**HIGH - #7**

**Issue:** `.env.example` shipped relative paths `MODEL_PATH=backend/models/...` that only resolve from repo root, while scripts run from `backend/`.

**Resolution:** FIXED - defaults now repo-relative to `backend/` (`models/risk_model.joblib`, `data/`).

**HIGH - #8**

**Issue:** MSG91 phone normalisation used `phone.replace('+91', '').replace('91', '')` which strips '91' ANYWHERE in the number (e.g. 9 1 9... mangling valid 10-digit digits).

**Resolution:** FIXED - anchored regex `re.sub(r'^\+?91', '', phone)`.

**3. MEDIUM / LOW Findings (all fixed)**

| # | Severity | Issue                                                                                                                                             | Fix                                                                                         |
|---|----------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| 1 | MEDIUM   | Escalation alerts created by <code>check_and_create_escalation_alerts</code> left <code>from_face</code> <code>to_recipient_count</code> set to 0 | Changed to <code>from_face=Courier</code> and <code>to_recipient_count=1</code>             |
| 2 | LOW      | <code>train_model.py</code> docstring referenced non-existent <code>&lt;font face=Courier&gt;scripts/start_for_output.py&lt;/font&gt;</code>      | Changed to <code>&lt;font face=Courier&gt;scripts/recompute_risk.py&lt;/font&gt;</code>     |
| 3 | LOW      | <code>startup.py</code> type annotation <code>&lt;font face=Courier&gt;list[tuple[str, str, str]]&lt;/font&gt;</code>                             | Changed to <code>&lt;font face=Courier&gt;list[tuple[str, object, str]]&lt;/font&gt;</code> |

**4. Verification After Remediation**

| Check                                     | Result                                                                                       |
|-------------------------------------------|----------------------------------------------------------------------------------------------|
| Alembic upgrade head (fresh DB)           | 4 migrations applied cleanly, head = 99f87406c0b8                                            |
| Seed / inventory / SWI / training         | users+zones seeded; 841 inventory events loaded; SWI computed; XGBoost (n=1441, holdout 25%) |
| AlertStatus.failed write                  | WROTE failed OK -> AlertStatus.failed 1 on migration-built DB                                |
| SOS lifecycle (mock SMS)                  | SOS #11: pending -> delivered (sms_sent=1, recipients=1, sent_at set) -> acknowledged OK     |
| pytest                                    | 5 passed                                                                                     |
| Smoke test (pristine DB)                  | 26/26 passed                                                                                 |
| Smoke test (warm DB, idempotent register) | 25/25 passed                                                                                 |

All critical, high, medium and low audit findings are resolved. No open errors remain. Out-of-scope environmental limits (no outbound network, no local Docker/Postgres, App Control DLL/exe policy) are noted but bypassed via fallbacks.